

#	Milestone	Generative AI Guidelines	Due	Mark Release
1	Background and Related Work	C: AI-Assisted Editing ¹	Oct. 07, Wed., AoE ²	Oct. 12, Mon.

This course is *project driven*. Students will work in teams to define, develop, and complete a project involving data modeling and representation over the course of one semester. The course project has two main objectives:

- Hands-on experience: Gain practical experience in analyzing and modeling real-world data, while developing skills in data representation, schema design, querying, implementation, evaluation, and teamwork.
- Research experience: Develop an understanding of the research *process*, from identifying a problem and reviewing existing approaches to designing, implementing, and evaluating different representations of data in the same domain.

Projects should aim to advance the state of the art, rather than simply reproduce an existing solution. This provides students with an opportunity to conduct meaningful project while completing the course requirements.

A data modeling project typically starts with a real-world domain and an information need. In this sense, a data modeling project may resemble a software or database project; however, there are important differences. A software project may primarily focus on implementing a functional system, while a data modeling project in this course focuses on understanding the structure and meaning of information and determining how that information can be represented, stored, and queried effectively for globally shared, accessible, and interoperable representations that can support Knowledge Graphs, the Giant Global Graph (GGG), and Linked Open Data³.

In this course, students will progressively explore different approaches to representing the same real-world domain. The project will begin with conceptual data modeling, including the identification of entities, attributes, and relationships and their representation using an Entity-Relationship Diagram (ERD). The conceptual model will then be transformed into a relational model (logical level of data modeling), including relational tables, keys, and constraints, and implemented and queried using SQL. The project will subsequently extend the representation toward semantic and knowledge representation, using RDF, RDFS, and OWL, and will explore the storage and querying of the resulting knowledge graph using a triple store and SPARQL:

Real-world domain → ERD → Relational Schema → SQL → RDF/OWL → Triple Store → SPARQL

The goal is not simply to implement these technologies independently. Rather, students should understand how different data models represent information and relationships, how one representation can be transformed into another, what information and semantics may be preserved or lost during such transformations, and how different representations affect the types of questions that can be answered.

Roughly, the data modeling project in this course consists of the following milestones (phases):

1. Project Definition & Team Formation (Proposal)
2. Background & Related Work (Literature Review)

¹ AI can be used to make improvements to the clarity or quality of student-created work to improve the final output, but no new content can be created using AI (e.g. language editing for lab reports). AI can be used, but your original work with no AI content must be provided in an appendix.

² <https://www.timeanddate.com/time/zones/aoe>

³ <https://www.dbpedia.org/>

3. Project Presentation: 3-Minute Talk (3MT)
4. Solution Design & Implementation (Data Modeling & Representation)
5. Evaluation (Experimentation)
6. Project Demonstration: 10-Minute Talk (10MT)

A project manual has been prepared to guide students through each milestone. Each milestone breaks the project into a series of smaller steps and deliverables. The current manual covers Milestone 2: Background and Related Work, also known as Literature Review.

1.1. Introduction

The purpose of this milestone is to understand how the domain of your project has been represented and modeled and to use that understanding to develop an initial data model for your project. Unlike a traditional literature review that focuses primarily on research methods, this milestone focuses on data modeling and representation. Your goal is to investigate how different systems, organizations, standards, or researchers have represented information in your chosen domain. You should answer the following questions:

- What information is important in this domain?
- How have existing systems or projects represented this information?
- What entities, attributes, and relationships appear in these representations?
- How do different data models differ?
- What are the strengths and limitations of these models?
- What can we learn from existing models when designing our own?
- How should your project domain be represented in your proposed data model?

The purpose of this milestone is therefore not simply to collect examples of existing databases or diagrams. Instead, you should develop an understanding of the structure of the domain and how that structure has been represented by others. This milestone concludes with a draft Entity-Relationship (ER) diagram for your project. This draft will serve as the starting point for the subsequent development of your data model and, ultimately, its representation as a knowledge graph. Think of this milestone as a journey from the domain to its data model:

Domain → Existing Representations → Comparison → Your Conceptual Model

1.2. History

You should begin with a brief historical review of the domain and the information it contains. The purpose is to provide context for a reader who is not familiar with the domain and to show how the need to represent and organize information has evolved. Trace the domain back to its earliest known roots, even if these roots predate computer systems. Look for the earliest practices, concepts, records, classifications, or information-management challenges related to your project.

You do *not* need to provide an exhaustive history. The goal is to give the reader a quick journey (odyssey) from the early origins of the domain to the way information is represented and managed today. For example, if your project concerns *“public-safety incidents”*, you might briefly discuss *“how incidents were historically recorded in reports or administrative records before the emergence of computerized information systems”*, followed by the development of structured databases and, more recently, interoperable and graph-based representations. Your historical review should answer, as appropriate:

- What is the earliest known form or root of the domain or information problem?
- How has the way information been recorded or organized evolved?
- When did computerized data management become relevant?
- What major developments led to the current ways of representing the data?

- Why have more structured and interoperable representations become important?

Keep it brief: This is a historical overview, not a history thesis. Focus on the major milestones that help the reader understand how the domain and its data representation evolved.

1.3. Hierarchy (Categories and Subcategories)

Search for existing databases, data models, schemas, standards, ontologies, knowledge graphs, APIs, or other structured representations relevant to your domain. Your sources may come from different communities:

- academic research projects;
- government or municipal data systems;
- industry databases;
- open-data portals;
- international standards;
- domain-specific information systems;
- relational databases;
- XML or JSON schemas;
- RDF/knowledge graphs;
- existing ontologies.

You should not assume that there is one correct way to model a domain. Different models may represent the same domain in different ways because they were designed for different purposes. For example, one system may represent “*Person* → *Incident* → *Location*” while another may additionally represent “*Person* → *Role* → *Incident* → *Organization* → *Location* → *Time*” Both models may be appropriate for their intended purposes. Your task is to identify these differences and understand why they exist. When examining an existing data model, identify:

- Entities: What things are represented?
- Attributes: What properties are recorded about them?
- Relationships: How are entities connected?
- Identifiers: How are entities uniquely identified?
- Constraints: What rules or restrictions are imposed?
- Temporal information: Is time represented?
- Location: Is geographic information represented?
- Provenance: Is the source of information represented?
- Granularity: How detailed is the representation?
- Interoperability: Can the data be connected to other systems?

You do not necessarily need to answer every question for every model. Focus on the characteristics that are relevant to your project. Existing representations should be organized into meaningful categories rather than presented as a sequence of unrelated examples. For example, depending on your domain, you might categorize models according to:

- purpose: operational vs. analytical;
- representation: relational vs. document-based vs. graph-based;
- scope: local vs. national vs. international;
- granularity: coarse-grained vs. fine-grained;
- source: government vs. academic vs. industry;
- modeling perspective: entity-centered vs. event-centered;
- or another perspective that makes sense for your domain.

For example:

“Existing representations of public-safety information can be categorized according to their primary purpose into operational incident databases, statistical/analytical datasets, and interoperable knowledge representations.”

You may then further divide one category according to another useful perspective. Usually, **two levels of categorization** are sufficient for a course project. The goal is not to construct an elaborate taxonomy. The goal is to help the reader understand the different ways in which the domain has been modeled. A simple hierarchy or taxonomy can be used to visualize the categories.

1.4. Comparative Analysis

After identifying representative models, compare them. For each important model, explain:

- What is the purpose of the model?
- What entities does it represent?
- What are the important relationships?
- What information does it capture well?
- What information does it omit or simplify?
- What are its strengths?
- What are its limitations?
- What aspects are relevant to your project?

Do not simply write:

“Model A contains Person and Incident. Model B contains Person and Event. Model C contains Person, Event, and Location”.

Instead, try to explain the modeling decisions:

“Model A treats an incident as the central entity and associates people and locations with the incident. Model B represents events more explicitly and distinguishes between the event itself and the people participating in it. This distinction allows Model B to represent multiple roles for the same person across different events.”

A comparison table can be very useful, but not a replacement for the written analysis.

Existing Model	Main Purpose	Key Entities	Key Relationships	Strengths	Limitations
Model A	...	...	...	...	...
Model B	...	...	...	...	...
Model C	...	...	...	...	...

	EAV	DT	OEAV	OCOM
Sparseness	No null values are stored	No null values are stored	No null values are stored	No null values are stored
Volatility	New attributes can be easily added by just adding the name in attribute column of EAV	One new table can be easily created for new attribute	New attributes can be easily added by just adding the name in Attribute dictionary and Index table and corresponding coded value in OEAV table	One new table can be easily created for new attribute
Handling Heterogeneous Data	One dedicated value column. So capable of handling one data type only	Handles heterogeneous data very easily by ensuring different table for different attributes	Stores only numeric data due to need of numeric code construction	Partially handles the homogeneous data by constructing equivalent numeric codes for data other than numeric.
Efficient Attribute Centric Query	Whole database needs to be searched.	Performs well (different tables for different attributes)	Performs moderately (index table exists for quick access of attribute but numeric codes need to be decoded)	Performs moderately (different tables for different attributes but numeric codes need to be decoded)
Efficient Entity Centric Query	Whole database needs to be searched	All tables need to be searched	Whole database need to be searched	All tables need to be searched
Calculation Overhead	No extra calculations required	No extra calculations required	Numeric codes need to be coded and decoded	Numeric codes need to be coded and decoded
Anomalies (Insertion, Modification & Deletion)	No problem while insertion, deletion and modification	No problem while insertion, deletion and modification	Sorting of OEAV table and reconstruction of index table is required	No problem while insertion, deletion and modification but codes need to be calculated
Machine Dependency	Easily portable to new machine	Easily portable to new machine	Reconstruction of index table is required	Easily portable to new machine

Table 1. Sample Comparison of various data models⁴.

1.5. Position Your Project and Draft ER Diagram

The final and most important deliverable of this milestone is a draft Entity-Relationship (ER) diagram for your project domain. Based on your investigation of the domain and existing data models, your team should propose an initial conceptual data model that addresses the gaps, limitations, or missing information identified in existing models. For example, if your project concerns *“public-safety incidents”*, your initial model might contain entities such as *“Person, Incident, Location, Organization”* with relationships such as

- *“Person participates in Incident”*
- *Incident occurs at Location*
- *Organization responds to Incident”*

Along with the diagram, briefly discuss the key modeling decisions and explain which gaps or limitations in existing data models your proposed model is intended to address. You do not need to provide a detailed justification for every element of the diagram; focus on the most important additions, refinements, or distinctions introduced by your model.

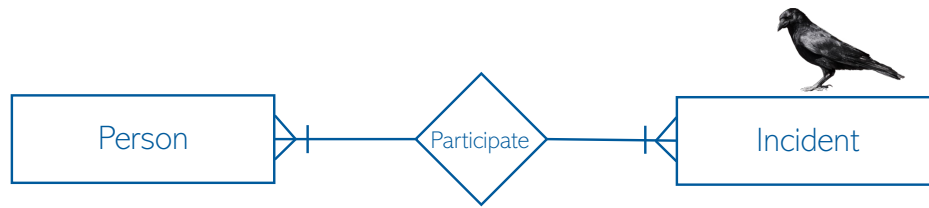
Important: Draft Means Draft. Your ER diagram at this stage is not expected to be final. The purpose of the draft is to demonstrate that your team has investigated the domain sufficiently to propose a reasonable initial representation. You will refine the model as you learn more about the domain and as you move toward knowledge-graph representation. You should therefore expect questions such as:

- Is this really an entity or should it be an attribute?
- Should these two entities be separate?
- Is this relationship many-to-many?
- Should this information be represented explicitly?
- Are we losing important information by modeling it this way?
- Can this model support the questions our project needs to answer?

From ER Model to Knowledge Graph. Because this course ultimately moves from data modeling toward knowledge graphs, your draft ER diagram should be designed with that transition in mind. At this stage, you are

⁴ <https://www.inderscienceonline.com/doi/abs/10.1504/IJCSE.2018.093777>

not required to convert the ER diagram into OWL. However, you should consider whether the concepts and relationships you identify could later be represented as entities and relationships in a knowledge graph.



may later become a graph relationship (triple) such as: “*Person* → *participatesIn* → *Incident*”. Similarly, important attributes or concepts may eventually become properties or additional entities in the knowledge graph. This does not mean that an ER diagram and a knowledge graph are the same thing. The purpose is to begin thinking about the domain in terms of entities, relationships, meaning, and interoperability.

1.6. Submission Guidelines

1.6.1. Deliverables

- Submission must be written in English, in the current ACM two-column conference format in LaTeX. Overleaf⁵ templates are available from the ACM Website⁶. Use the *sigconf* proceedings template.
- The submission must be *exactly* 2 pages (text) + 1 page (ER Diagram) in length, no more and no less, including author list, figures, tables, and references, with unlimited pages for C: AI-Assisted Editing appendix, and must be authored by all team members, with the graduate student listed last in the author list. We understand that fitting all the required content into exactly one page can be challenging. However, this is intentional: an important part of communication is learning how to retouch, refine, condense, and prioritize information without losing its essential meaning. We recommend initially writing a longer version (e.g., 5-10 pages) and then gradually reducing and retouching it until you can effectively communicate the key ideas within the required three-page limit.
- The submission must be a single zip file named: 2_Background_firstname1_firstname2 [...].zip, e.g., 2_Background_Kovvy_Hossein.zip, containing:
 - o The LaTeX source files and any required supplementary files;
 - o The compiled three-page PDF file.

A *sample* is attached to this manual and is also available [here](#)⁷ (2_Background.tex).

1.6.2. Team and Individual Contributions

- Each team must maintain a Git-based repository (e.g., GitLab, GitHub, ...) ⁸ for the project. The repository must be associated with the individual accounts of all team members. The repository will be used to document the development of the project and to make individual contributions traceable. The repository should be used throughout the project rather than only immediately before a submission deadline. The instructor and/or marker may consider the repository’s commit history, pull requests, issues, documentation, and other project records when assess individual contributions. The number of commits alone will not be used as a measure of contribution. The quality, significance, and nature of each member's contributions will also be considered.

⁵ <https://www.overleaf.com/gallery/tagged/acm-official>

⁶ <https://www.acm.org/publications/proceedings-template>

⁷ <https://www.overleaf.com/read/bsrxtwqgggsb>

⁸ Web-based applications and cloud services may collect, use and/or disclose your personal information and store or access that information outside of Canada. Personal information includes, but not limited to, identifiable individual’s information such as your name or your email address. At UVic, GitLab at gitlab.uvic.ca is highly recommended for coursework, research and development projects. For more information, see <https://www.uvic.ca/systems/services/pages/gitlab.php>. At your team’s discretion, you may use GitHub to store information about your project, but we do not require nor recommend storing any personal identifying information. Also, we do not require public repo, but it should be accessible to instructors and markers for evaluation purposes.

- All team members are expected to make substantive contributions, and no member should be limited solely to coordination or project management.

1.6.3. Marking Scheme

- 10% Abstract
- 10% Introduction
- 10% History
- 10% Hierarchy + Figure
- 20% Comparative Analysis + Table
- 30% ERD + Position your project (what and how existing gaps would be addressed).
- 10% Author list, style, format, repo, file naming convention, ...